

Answer **all** questions.

1. (a) Delyth has been given the task of constructing a resistor of resistance $56\ \Omega$ using wire labelled: **diameter** $0.15\ \text{mm}$, resistivity $4.9 \times 10^{-7}\ \Omega\ \text{m}$. Calculate the length of wire that she should use. [3]

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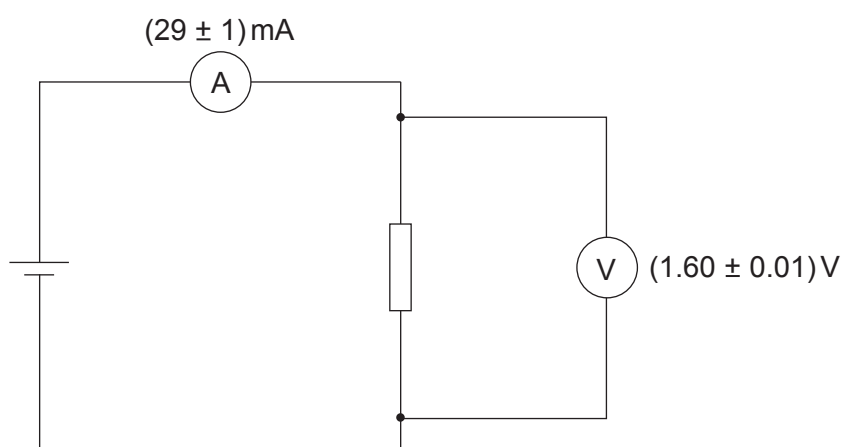
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- (b) Delyth tests this length of wire by placing it in the circuit shown.



Delyth concludes from the meter readings, shown on the diagram, that her resistor does **not** have a resistance of $56\ \Omega$. Evaluate whether or not this is a valid conclusion. [3]

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